CS348 Computer Networks Quiz-2, Fri 12 Sep 2025, 08.35am-09.20am, Max. Marks: 10

Name: ____________________________ Roll number: _______________ House: ____________________

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only.
- All answers have to be (briefly) explained. State any necessary assumptions.

1. **[2+1+2=5 marks]** A sysad Mnzncrfe Cdconisn has to manage 4 departments D0, D1, D2, D3, which need to support 500, 500, 100 and 100 hosts respectively. She got her ISP to give her some blocks of addresses, specifically: 194.1.0.*, 194.1.1.*, 194.1.2.*, 194.1.3.*, 194.1.4.*, 194.1.5.*

   (a) Help Cdconisn allocate the above IP prefixes to the 4 departments so as to minimize the number of entries in the routers within her institute? Specify the subnet(s) of each department. Allocate so that she can return one of the above 6 blocks, to save some money for the institute.

   (b) Can all communication within department D0 happen without a layer3 router? Justify (< 30 words).

   (c) Cdconisn heard about CIDR and wants to know how the other routers in the Internet will represent her company in their routing table. Help her. (Max 3-4 lines).

2. **[5 marks]** Prof Likbasr Canamgal deploys the bridged ethernet below with 5 learning bridges. However the professor installed a buggy code on B2 due to which it does not learn anything. They all start with empty forwarding tables. Draw the forwarding tables at each of the 4 non-buggy bridges, after each of the following packet transmissions reach their respective receivers. A total of 12 tables – draw in pencil, organize neatly! Explain briefly. You may denote the two interfaces of each bridge as *top-T* and *bottom-B* respectively (this notation is not marked in the diagram).
Transmission-1: N1 to N4, Transmission-2: N3 to N1, Transmission-3: N4 to N3